

Converting Between Exponential and Logarithmic Forms

Rewriting an exponential statement as a logarithm, rewriting a logarithm as an exponential statement, evaluating logarithms by converting, changing base, and solving equations by switching forms

The one idea behind every problem. A logarithm is an exponent. The two forms below say exactly the same thing about the same three numbers — only the question changes. Exponential form asks “what is the result?”; logarithmic form asks “what is the exponent?”

$$b^y = x \quad \Longleftrightarrow \quad \log_b x = y$$

base b stays the base in both forms

exponent y is what the logarithm equals

result x is what the logarithm is taken of

Reference card. No values are filled in — use the numbers given in each problem.

Directions.

- Before converting, label which given number is the base, which is the exponent, and which is the result.
- Give exact answers unless the problem asks for a decimal; then round to three decimal places.
- Write your final answer on the answer line.

1. The statement $5^3 = 125$ is in exponential form.

(a) Name the base, the exponent, and the result.

(b) Rewrite the statement in logarithmic form.

(c) What number does $\log_5 125$ equal?

Answer: _____

- 2.** The statement $\log_2 32 = 5$ is in logarithmic form.
- (a) Which number is the base, and which one is the exponent?
- (b) Rewrite the statement in exponential form.
- (c) Evaluate the power you wrote in part (b).

Answer: _____

- 3.** Evaluate $\log_3 81$ without a calculator.
- (a) Fill in the table of powers.

power	3^1	3^2	3^3	3^4
value				

- (b) $\log_3 81$ asks which of those exponents produces 81. Write the exponential statement, then the value of the logarithm.

Answer: _____

4. A logarithm can be negative, because an exponent can be negative.

Consider the exponential statement $10^{-2} = 0.01$.

(a) Rewrite it in logarithmic form.

(b) What is $\log_{10} 0.01$?

(c) In one sentence, say why the logarithm of a number between 0 and 1 comes out negative when the base is bigger than 1.

Answer: _____

5. Solve $2^x = 40$.

(a) The unknown is the exponent, so the logarithmic form isolates it. Rewrite the equation in logarithmic form.

(b) Between which two consecutive whole numbers must x lie? Say how you know from powers of 2.

(c) Give x to three decimal places.

Answer: _____

6. Evaluate $\log_8 2$ exactly.

(a) Write the conversion: $\log_8 2 = y$ becomes what exponential statement?

(b) 2 is smaller than the base 8, so the exponent is a fraction between 0 and 1. Find it exactly, as a fraction in lowest terms.

Answer: _____

7. Solve $\log_4 x = \frac{3}{2}$.

(a) Here the unknown sits inside the logarithm, so exponential form is the one that isolates it. Rewrite the equation.

(b) Evaluate the power. A fractional exponent is a root and a power: the denominator is the root.

(c) Check your answer by substituting it back into the original logarithmic statement.

Answer: _____

8. Most calculators only have $\ln$ and $\log_{10}$, so a logarithm in another base has to be rewritten.

(a) Starting from $\log_3 x = y$, convert to exponential form, take the natural logarithm of both sides, and solve for y . Write $\log_3 x$ as a quotient of natural logarithms.

(b) Use your formula to approximate $\log_3 20$ to three decimal places.

(c) Write both your part (a) expression and your part (b) decimal on the answer line.

Answer: _____

9. Solve $\log_5(2x - 3) = 2$.

(a) Convert to exponential form. The entire expression $2x - 3$ is what the logarithm was taken of, so it becomes the result.

(b) Solve the equation that remains.

(c) Verify that your value makes $2x - 3$ positive, and say in one sentence why that check is required.

Answer: _____

10. Challenge. An investment of 1200 dollars grows by 4.5 percent each year, so its value after t years is $1200 \cdot 1.045^t$.

- (a) Write the equation that says the investment has doubled.
- (b) Divide both sides by the starting amount, then convert the result to logarithmic form to isolate t .
- (c) Find the doubling time to three decimal places.

Answer: _____